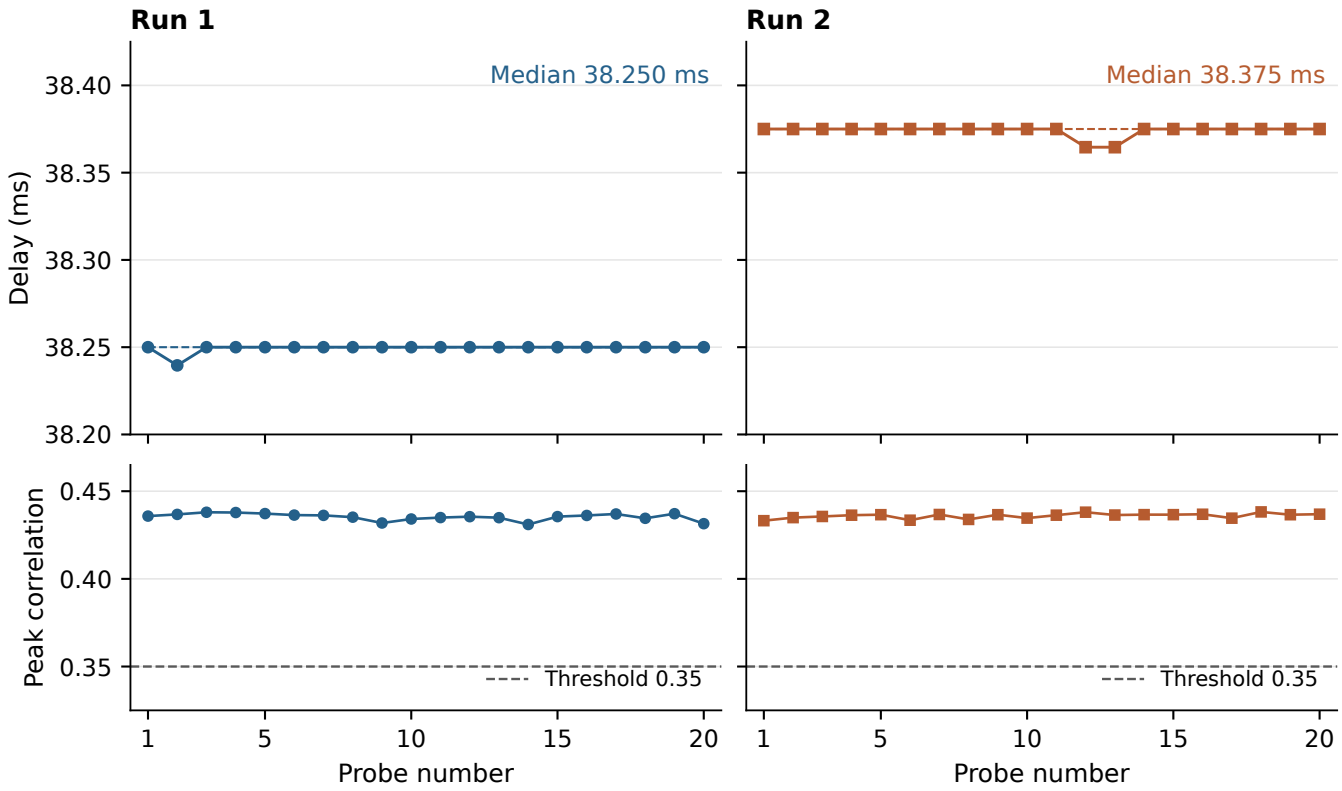
Playback-to-microphone delay

Audient iD14 · Edifier speaker · Earthworks M50

Run 1 median	Run 2 median	Between-run shift
38.250 ms	**38.375 ms**	**0.125 ms**

Recorded at 96 kHz, with approximately 10 cm microphone separation (operator reported). Each separately opened stream provided 20 accepted sweeps.



Ranges: Run 1, 38.2396–38.2500 ms; Run 2, 38.3646–38.3750 ms. Each run's peak-to-peak spread was one sample (0.0104 ms). The strength threshold was 0.35; probes were 100 Hz–20 kHz sweeps.

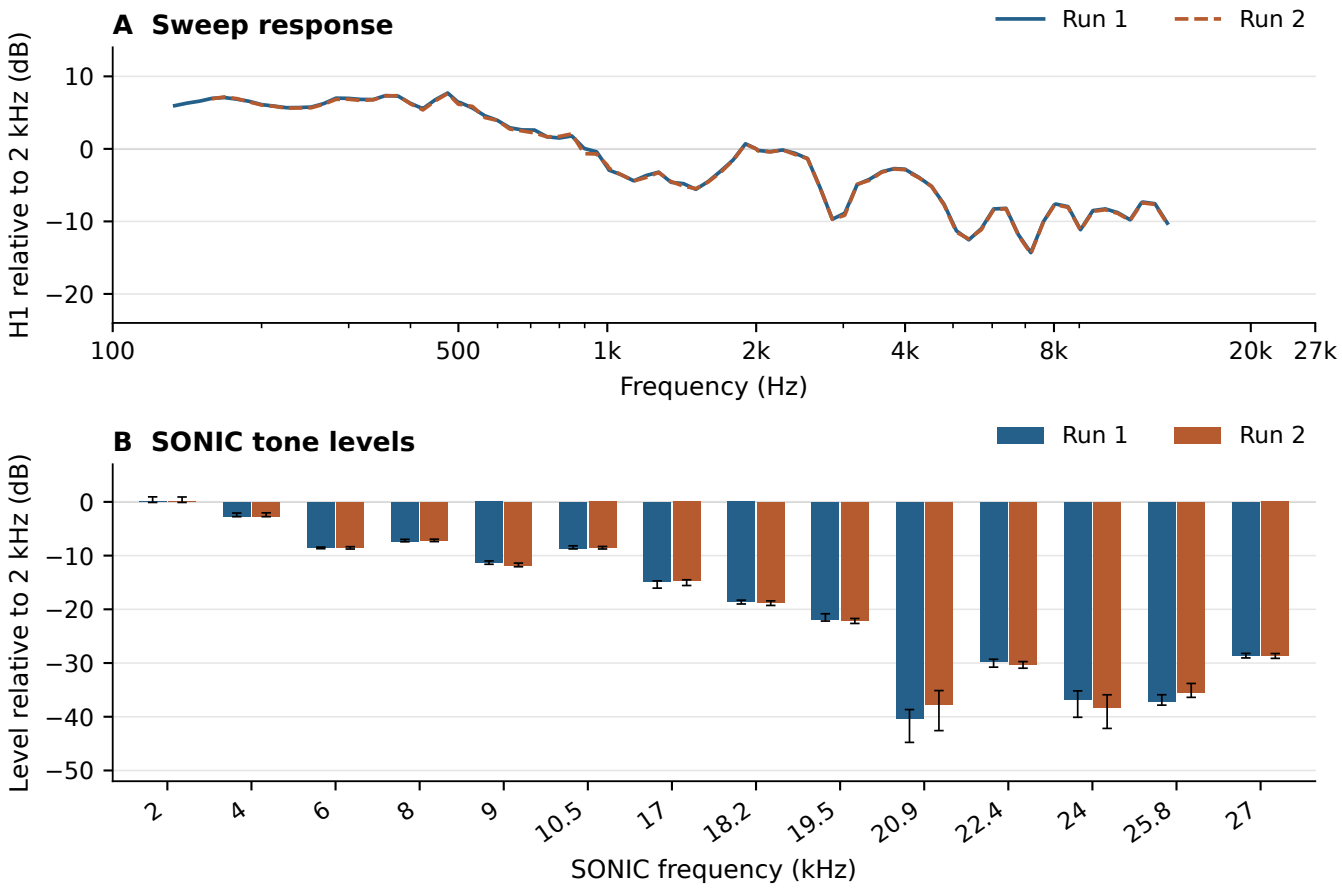
Interpretation. Each run was stable over its 20 recorded probes, but the baselines differed. The 0.125 ms median shift is between-run variation; more stream starts are needed to characterize it. This application roundtrip includes buffering and the acoustic path. Within-run spread does not establish calibrated accuracy, isolated speaker delay or neural latency.

Recorded frequency levels

SONIC tone reconstruction · 14 frequencies · 2–27 kHz

Tone duration	Ramp duration	Recorded pips
10 ms	**2 ms**	**16,800**

Locally reconstructed Sheep E tones in two 84-second blocks.



Gaps mark unsupported sweep bands. Tone bars show medians; whiskers are 5th–95th percentiles of 600 pips per frequency per recording, not confidence intervals.

Finding. All 14 components appear in both recordings at unequal levels. Relative to 2 kHz, 17–19.5 kHz was 15–22 dB lower; 20.9–25.8 kHz was 30–40 dB lower. These uncorrected full-chain measurements leave SPL, speaker-only response and distortion unqualified.

Next. Fix the geometry, calibrate microphone/SPL, test separated tones, and validate timing with an independent electrical reference.